



Jayesh Sonawane

Roll No.:210106027

B.Tech - Biosciences and Bioengineering
Indian Institute Of Technology, Guwahati

+91-8698547585

piephi367@gmail.com

j.sonawane@iitg.ac.in

GitHub | LinkedIn

EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech	Indian Institute of Technology, Guwahati	7.59 (Current)	2021-Present
Senior Secondary	Maharashtra State Board	91.0%	2021
Secondary	Maharashtra State Board	90.0%	2019

EXPERIENCE

- **Deepsoft Technologies** Mar 2024 - May 2024
Data Science Intern Remote
 - Developed a pipeline to **recover prompts** from AI-generated text using the **Mistral-7B-Instruct** model.
 - Fine-tuned the model with **LoRA**, achieved **0.44 ROUGE-L** score which is **12%** better than **Few-shot ICL**.
- **UiT- The Arctic University of Norway** Jan 2024 - Mar 2024
Machine Learning Intern Under Dr.Anowarul Habib Remote
 - Enhanced an innovative library for **reconstructing image data** from sparse wavefield signals of piezoelectric sensors.
 - Attained an **MSE of 0.69** on **200** optimal sensors using **19** training examples through optimal sensor placement.
 - Used **ResNet50** for damage detection obtaining **96.54% accuracy** and created a novel approach for localizing damage.
- **University of New South Wales** Dec 2022 - May 2023
NLP Research Intern Under Dr.Rohitash Chandra Publication
 - Trained **BERT** on the **SenWave** dataset of tweets for multi-label sentiment classification in **11** categories.
 - Performed detailed analysis of the **Global COVID-19 Twitter Dataset**, specifically targeting anti-vaxxer sentiments.
 - Identified a **strong correlation** between tweet volume, COVID-19 case numbers, and mean sentiment polarity scores.
 - Observed **normalization** of sentiment polarity scores towards 0 post-vaccine rollout.

PROJECTS

- **Adobe Behavior Simulation Challenge** Nov. 2023 - Dec. 2023
Inter IIT Tech Meet 12.0, IIT Madras GitHub
 - Predicted Twitter post likes and generated text content based on given tweets.
 - Generated image captions with **BLIP-2** and categorized content using **DistilBERT**, achieving **74.3%** accuracy.
 - Applied regressors within each bucket to predict likes achieving **RMSE of 3812.6**.
 - Fine-tuned **Llama-2** using **PEFT**, achieving a **BLEU score of 0.004** for content generation.
- **Devrev's AI Agent 007** Nov. 2023 - Dec. 2023
Inter IIT Tech Meet 12.0, IIT Madras GitHub
 - Developed an LLM chatbot with a **dynamically adjustable toolset** for accurate query responses.
 - Used **Retrieval-ICL** for few-shot retrieval, and **RAG** for dynamic retrieval.
 - Reduced hallucinations using **corrective re-prompting** and **COT** prompting, enhancing accuracy by **30%**.
 - Explored Human-guided query generation and augmentations for **synthetic dataset creation**.
 - Attained **85%** accuracy, tested on **1000** queries varying complexity with **7.05 s.** average latency.
- **Cloudphysician's The Vital Extraction Challenge** Jan. 2023 - Feb. 2023
Inter IIT Tech Meet 11.0, IIT Kanpur GitHub
 - Extraction of **important vitals** like Heart Rate, SPO2, RR, etc. from **images of ECG Machines**.
 - Used **UNET++** for screen extraction, and **PaddleOCR** for number detection.
 - Developed novel **CRABNet** architecture for mapping extracted numbers with their classes.
 - Achieved an accuracy of **89.5%** with an inference time of **8 s** and Extracted **HR graph** using **OpenCV** techniques.

ACHIEVEMENTS

- **Kaggle Notebook Master**, Best rank of **505** among 20 million users on the Kaggle platform.
- **Amazon ML Summer School 2024**, Selected in the top **3000** candidates to attend the ML Research program at Amazon.
- **Adobe Behavior Simulation Challenge**, Secured the **7th** Position out of 22 participating IITs.
- **Codeforces**, rating of **1200** demonstrating proficiency in competitive programming.

TECHNICAL SKILLS

- **Programming:** Python, C/C++, R, SQL
- **Tools/Frameworks:** PyTorch, Tensorflow, Keras, OpenCV, Numpy, Pandas, Matplotlib, Seaborn, PySpark
- **Database Management:** MySQL, MongoDB

KEY COURSES TAKEN

- **Mathematics:** Linear Algebra, Probability, Statistics, Multivariable Calculus
- **Computer Science:** Introduction to Computing, Data Structures and Algorithms, Recommendation System Design Using Deep Learning, Pattern Recognition and Machine Learning
- **Online MOOCs(Coursera):** Machine Learning, Deep Learning, Sequence Models, Convolutional Neural Networks, Generative AI with Langchain and Huggingface, Data Analysis with Python, Prompt Engineering